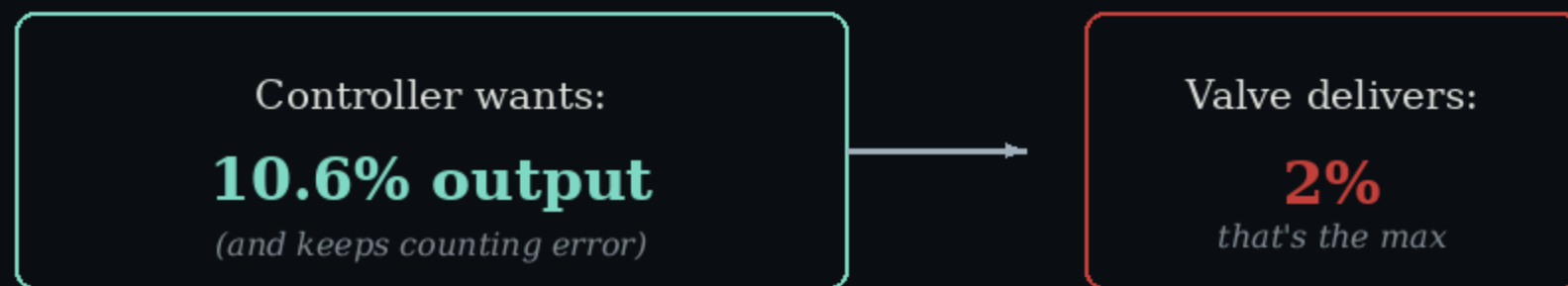




A QUICK ENGINEERING BREATH

THE INTEGRAL DOESN'T KNOW

The controller can ask for anything. The actuator can only give what it physically has.



A naive integral term doesn't know the difference. It keeps accumulating error the whole time it's saturated — building a hidden debt that has to unwind later.

What that debt actually costs — next.

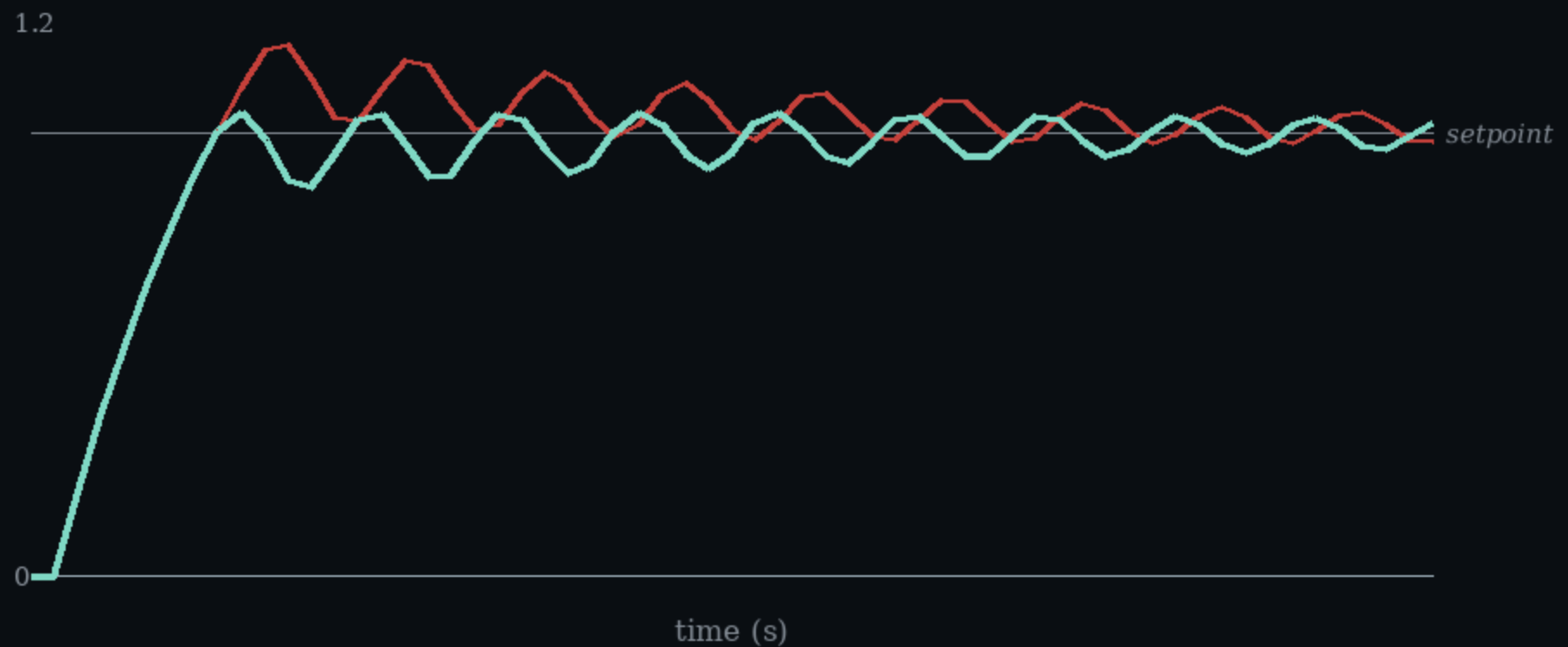


A QUICK ENGINEERING BREATH

SAME LOOP, ONE LINE OF LOGIC

Identical process, identical limits — the only difference is whether the integral knows it's saturated.

— without anti-windup — ~20% overshoot — with anti-windup — ~5% overshoot



Same tuning. Same limits. One line of logic is the entire difference.

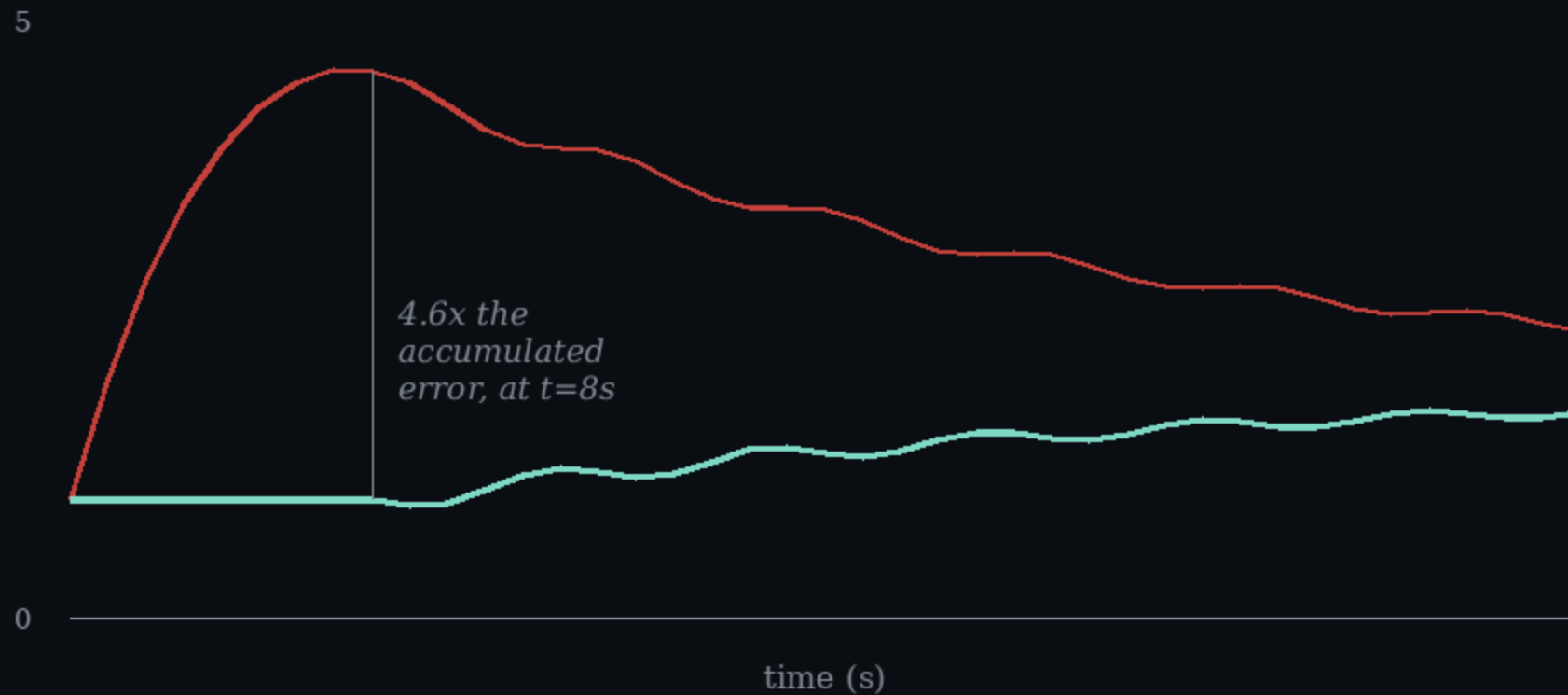


A QUICK ENGINEERING BREATH

THE HIDDEN DEBT, MADE VISIBLE

The integral accumulator itself — what the controller "remembers" as unresolved error.

— accumulated error, unprotected — accumulated error, protected



That's the debt the overshoot on the last slide was paying off.